

This assignment contributes 10% towards your final grade. It should be submitted by 3pm on Thursday 23 April 2026. In the absence of an approved extension, assignments submitted after the due date will attract a penalty as outlined in the course profile. Prepare your assignment as a **pdf file**, either by typing it, writing on a tablet or by scanning/photographing your handwritten work. Ensure that your name appears on the first page of your submission. Check that your pdf file is legible and that the file size is not excessive. Files that are poorly scanned and/or illegible may not be marked. Use of generative AI is permitted, but not encouraged for this assessment. If you choose to use AI, you must do so appropriately and provide clear acknowledgement and referencing in line with UQ's guidelines.

1. (8 marks) You find out that a 4-stage LFSR stream cipher $\mathcal{C}$ was used to encrypt the plaintext $P = 101101011101$ to obtain the ciphertext $C = 110001110011$.

- (a) Use this information to determine the recurrence relation and first keystream bits used to construct $\mathcal{C}$.
- (b) What is the period of $\mathcal{C}$?
- (c) Assume that an m -stage LFSR has maximal period when initialised with keystream bits $(\ell_1, \dots, \ell_m)$. Will the same LFSR have maximal period if initialised on a different (and non-zero) sequence of keystream bits $(\ell'_1, \dots, \ell'_m)$? Prove it or find a counterexample.

2. (10 marks) Consider the 3-stage Feistel cipher, taking as input a 12-bit plaintext. The master key is

$$K = (13)(5264),$$

and each subkey K_i , $i = 1, 2, 3$ is the permutation obtained by applying K i times.

Note: recall we are using the convention according to which $K \cdot (\text{abcdef}) = (\text{ceafdb})$.

- (a) Determine at least one plaintext that is encrypted to itself by this Feistel cipher.
- (b) Given the plaintext 2748, convert it to binary, encrypt it using the Feistel cipher and determine the corresponding ciphertext. (Show all your working.)

3. (10 marks) Recall that in AES computations are performed in the field

$$\text{GF}(2^8) \cong \mathbb{Z}_2[x] / (x^8 + x^4 + x^3 + x + 1)$$

- (a) Determine the bitstring and polynomial representations of the field elements

$$\{7b\}, \{c3\}, \{7b\} \oplus \{c3\}, \{7b\} \cdot \{c3\}, \{7b\}^{-1}.$$

(Show your working.)

- (b) Compute the image of $\{7b\}$ through the AES substitution box. (Show your working.)

4. (10 marks) Consider the permutation

$$\pi = (0\ 7)(1\ b\ c\ 8)(2\ 3\ 9\ a)(4\ f\ 5\ e\ 6\ d)$$

of the hexadecimal number system.

- (a) Construct the associated S -box, and determine the bias of $X_2 \oplus X_3 \oplus Y_2 \oplus Y_3$ and $X_1 \oplus X_2 \oplus Y_1 \oplus Y_4$.
- (b) Determine the most frequent differential $(\Delta X, \Delta Y)$ such that $\Delta X = (1011)$.

5. (12 marks) Denote by E_p the elliptic curve $y^2 = x^3 + 2x + 1$ over the field $\mathbb{F}_p$.

- (a) Determine at least one value of p such that E_p is singular.
- (b) Determine the two points P, Q belonging to E_7 such that the x -coordinate of P, Q is zero.
- (c) For points P and Q from part (b), prove that $P = -Q$.
- (d) What is the group structure on E_7 ?
- (e) (Feel free to use online tools or the code on Blackboard for this part.) Consider the Diffie–Hellman key exchange protocol for E_{109} , with generator $G = (3, 19)$. Suppose Alice chooses her secret parameter to be 7 and Bob chooses his secret parameter to be 15. Compute their resulting private Diffie-Hellman key.